VIT BHOPAL UNIVERSITY

REPORT

UNIT COVERTER PROJECT

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UNIT CONVERTER PROJECT REPORT

1. Introduction

The Unit Converter is a simple Python-based project that allows users to convert values between different units such as length, weight, temperature, and time. This project is designed for beginners to understand basic programming concepts like functions, loops, and conditional statements.

2. Objective

The main objective of this project is:

- To perform unit conversions easily
- To understand Python basics like functions and loops
- To build a user-friendly console-based application

3. Features

- Converts Length (meter, kilometer, centimeter)
- Converts Weight (kilogram, gram)
- Converts Temperature (Celsius, Fahrenheit)
- Converts Time (seconds, minutes, hours, days)
- Menu-driven program
- Option to continue or exit

4. Technologies Used

- Programming Language: Python
- Platform: VS Code / Any Python IDE

5. Working of the Project

The program uses a loop (while True) to display a menu repeatedly. Based on user input:

- It calls the respective function (length, weight, temperature, time)
- Performs the calculation using formulas
- Displays the result

After each operation, the user is asked whether they want to continue or exit.

6. Code Structure

- main.py → Main menu and program flow
- length.py → Length conversion functions

- weight.py → Weight conversion functions
- temperature.py → Temperature conversion functions
- time_converter.py → Time conversion functions

7. Sample Output

===== UNIT CONVERTER =====

1. Length
2. Weight
3. Temperature
4. Time
5. Exit

Enter your choice: 4
Enter value: 450
Result: 7.5 hours

8. Conclusion

This project helped in understanding basic Python programming concepts. It is a useful and practical application that can be improved further by adding more units or creating a graphical user interface.

9. Future Scope

- Add more unit categories (speed, volume)
- Add GUI using Tkinter
- Store conversion history